# Worksheet - Tides

Try Q 1 - 7 below.

## UNDERSTANDING AND INQUIRING

### REMEMBER

1 What is the major cause of tides on Earth?

2 Why are there two high tides and two low tides in a day?

3 Even though the sun is much larger than the moon, it has much less effect on the tides. Why?

### THINK

4 Explain why the highest tides occur during a new moon or a full moon.

5 If the moon did not exist, would there still be tides? If so, how would they be different?

6 How would the tides be different if the Earth did not rotate? Explain your answer with diagrams showing the Earth, sun and moon.

7 On any given day, one high tide is higher than the other one. Suggest a reason for this.

### DRAW

8 Draw a diagram that shows one arrangement of the sun, moon and Earth that would cause a neap tide on the side of the Earth closest to the sun.

### CREATE

9 Design and act out a role-play that shows how the orbits of the Earth around the sun and the moon around the Earth cause tides. The role-play should involve at least four people, including a narrator to provide a commentary.

eBook plus

10 Use the **Tides** interactivity in your eBookPLUS to watch how changing the positions of the sun and the moon affects the tides on Earth. **int-0225**

work sheet 6.2 Surf's up